

TripMate: Phase 2 Progress Notes

Progress update, screenshots and repository link

Group No.: _____ · Course: Generative AI (PSIS activity) · Activity 2: Tool-Augmented / Agentic LLM Application

Submission date: 23 September 2026

Repository: <https://github.com/ananydoneria/tripmate> · Notebook: `TripMate_Agent.ipynb`

1. What works now

Part	Status	Evidence
6 tools (geocoding, weather, currency, holidays, Wikivoyage guide, budget)	Working	Tested against the live APIs; 23 offline unit tests with mocked HTTP
Agent: <code>create_agent</code> + Gemini tool calling	Working	Multi-tool plans (Jaipur: 8 tool calls in one turn)
System <code>PromptTemplate</code> with grounding and failure rules	Working	Rendered per call with today's date; relative dates resolved correctly
Memory (LangGraph checkpointing)	Working	Follow-up turns reuse city and dates
<code>PydanticOutputParser</code> itinerary chain with repair fallback	Working	Valid JSON for 3-day plans; repair path unit-tested with a fake LLM
Failure handling: outages, 429, HTTP 204, unknown places	Working	Outage simulation switch; structured <code>{ok: false, hint}</code> results
Streamlit UI and CLI	Working, polishing	Tool-trace panel, outage switches
Live evaluation (10 scenarios), report, video	In progress	Due in Phase 3

2. Output of the partially working system

```
... tripmate: python cli.py

TripMate - travel planning agent. Try: 'Plan 2 days in Udaipur next weekend'.
Commands: /plan <request>, /outage <service[off]>, /reset, /quit

You: How much is 2,000 UAE dirhams in Indian rupees?
    tool 1. convert_currency({"amount": 2000, "from_currency": "AED", "to_currency": "INR"}) -> ok

TripMate: 2,000 UAE dirhams (AED) is equivalent to approximately **52,297.73 Indian rupees (INR)** (at an exchange rate of 1
AED = 26.15 INR).

*Sources & caveats: Currency conversion is based on today's reference rate (September 17, 2026) provided by ExchangeRate-API
open access.*

[2 model calls, 4232 tokens, 60.22s]

You:
```

Figure 1. Real CLI session: the agent calls `convert_currency`; ECB rates lack AED, so the tool switches to the fallback provider and the answer reports the exact converted amount.

3. Problems found so far and how we handled them

- **Wrong place, no error.** The geocoder ranks "Manali" in Tamil Nadu (population 35k) above the Himalayan hill station (8k). Tools now return the resolved place and the other matches, accept qualified names ("Manali, Himachal Pradesh"), and the prompt tells the LLM to check the place and retry.
- **Qualifier bug.** "Jaipur, Rajasthan, India" failed because only the first qualifier was compared. Found in a live run where the agent recovered by retrying with "Jaipur, India"; fixed so every qualifier must match.
- **Gemini free-tier quota.** HTTP 429 after 5 requests/minute on gemini-2.5-flash and after 20 requests/day on gemini-3.5-flash. Added LangChain's `InMemoryRateLimiter` (4/min per model) and a fallback middleware that switches to gemini-3.5-flash-lite and skips the exhausted model for 10 minutes.
- **Unexpected API responses.** Wikivoyage rate-limits with a non-JSON 429 (`Retry-After: 21`): honoured and cached. Nager.Date answers HTTP 204 for India: reported as `no_data`, never as "no holidays". Forecasts stop at 16 days: last year's observations are used, labelled `historical_proxy`.

4. Next steps (until 30 September)

- Run and analyse the 10 live test scenarios; add grounding checks to the report.
- Finish the architecture and LLM-tool sequence diagrams, README and technical report.
- Record the 5-7 minute demo video; incorporate the Phase 2 feedback.